

Lab Class NLP:IX

Exercise 1 : Word Mover Distance

The lecture introduced the Word Mover's Distance (WMD) for measuring word vector similarity. WMD finds the minimum cumulative transportation cost to move all words from one sentence to words in another sentence in an embedding space [NLP:III-169]:

$$\text{WMD}(x, y) = \min_{\mathbf{T} \geq 0} \sum_{i,j=1}^n \mathbf{T}_{t_i, t_j} c(\mathbf{t}_i, \mathbf{t}_j)$$

where n is the number of words in the sentences and $\mathbf{T}$ is the transportation matrix and $c(\mathbf{t}_i, \mathbf{t}_j)$ is the cost between word embeddings $\mathbf{t}_i$ and $\mathbf{t}_j$. The cost is defined as the Euclidean distance between the word embeddings [NLP:III-161]:

$$c(\mathbf{t}_i, \mathbf{t}_j) = \|\mathbf{t}_i - \mathbf{t}_j\|_2 = \sqrt{\sum_{d=1}^p (t_{i,d} - t_{j,d})^2}$$

You are given the sentences A, B, and C and the 3-dimensional word vectors $[d_1, d_2, d_3]$ for all occurring words:

- Sentence A: "The **cat** **climbed** the tree."
- Sentence B: "The **feline** **scaled** the tree."
- Sentence C: "The **kitten** **ascended** the tree."

	the	cat	climbed	tree	feline	scaled	kitten	ascended
d_1	0.1	0.4	0.7	1.0	0.35	0.75	0.45	0.77
d_2	0.2	0.5	0.8	1.1	0.55	0.85	0.57	0.87
d_3	0.3	0.6	0.9	1.2	0.65	0.95	0.67	0.97

Answer

In a real implementation (e.g., using linear programming or a network simplex algorithm), you (1) compute **every pairwise distance** between words in the two sentences, then (2) solve the optimization over all pairwise flows simultaneously. The minimum is **not** found **greedily** by taking the nearest neighbor for each word independently.

If the sentences have different lengths (or, more generally, different normalized bag-of-words distributions), the WMD optimizer may split the transport flow from one word across multiple words or combine flow from multiple words into one word. E.g., sentence A has one word and sentence B has two words that are similar to it. Then the word in A has a mass of 1.0, and the two words in B have a mass of 0.5 each. The optimizer may split the flow from the word in A into two flows of 0.5 each to the two words in B, if that minimizes the total cost.

- (a) Calculate the WMD between Sentence A and Sentence B.

Answer

1P (-.25P for each small calculation error)

Pairwise distances between Sentence A and Sentence B:

$$\begin{aligned}d(\text{the}, \text{the}) &= \sqrt{(0.1 - 0.1)^2 + (0.2 - 0.2)^2 + (0.3 - 0.3)^2} = 0.0000 \\d(\text{cat}, \text{feline}) &= \sqrt{(0.4 - 0.35)^2 + (0.5 - 0.55)^2 + (0.6 - 0.65)^2} = 0.0866 \\d(\text{climbed}, \text{scaled}) &= \sqrt{(0.7 - 0.75)^2 + (0.8 - 0.85)^2 + (0.9 - 0.95)^2} = 0.0866 \\d(\text{tree}, \text{tree}) &= \sqrt{(1.0 - 1.0)^2 + (1.1 - 1.1)^2 + (1.2 - 1.2)^2} = 0.0000\end{aligned}$$

WMD calculation between Sentence A and Sentence B:

$$\begin{aligned}\text{WMD}(A, B) &= d(\text{the}, \text{the}) + d(\text{cat}, \text{feline}) + d(\text{climbed}, \text{scaled}) + d(\text{the}, \text{the}) + d(\text{tree}, \text{tree}) \\&= 0 + 0.0866 + 0.0866 + 0 + 0 = 0.1732\end{aligned}$$

(b) Calculate the WMD between Sentence A and Sentence C.

Answer

1P (-.25P for each small calculation error)

Pairwise distances between Sentence A and Sentence C:

$$\begin{aligned}d(\text{the}, \text{the}) &= \sqrt{(0.1 - 0.1)^2 + (0.2 - 0.2)^2 + (0.3 - 0.3)^2} = 0.0000 \\d(\text{cat}, \text{kitten}) &= \sqrt{(0.4 - 0.45)^2 + (0.5 - 0.57)^2 + (0.6 - 0.67)^2} = 0.1109 \\d(\text{climbed}, \text{ascended}) &= \sqrt{(0.7 - 0.77)^2 + (0.8 - 0.87)^2 + (0.9 - 0.97)^2} = 0.1212 \\d(\text{tree}, \text{tree}) &= \sqrt{(1.0 - 1.0)^2 + (1.1 - 1.1)^2 + (1.2 - 1.2)^2} = 0.0000\end{aligned}$$

WMD calculation between Sentence A and Sentence C:

$$\begin{aligned}\text{WMD}(A, C) &= d(\text{the}, \text{the}) + d(\text{cat}, \text{kitten}) + d(\text{climbed}, \text{ascended}) + d(\text{the}, \text{the}) + d(\text{tree}, \text{tree}) \\&= 0 + 0.1109 + 0.1212 + 0 + 0 = 0.2321\end{aligned}$$

(c) Which sentence B or C is more similar to Sentence A?

Answer

1P $\text{WMD}(A, B) < \text{WMD}(A, C)$. Therefore, Sentence B is more similar to Sentence A.

Exercise 2 : Sentence Embeddings

The lecture introduced sentence embeddings as a way to represent sentences in a continuous vector space. One approach to generating sentence embeddings is to average the word embeddings $\mathbf{w}$ of all words in a sentence s :

$$\mathbf{s}_{emb} = \frac{1}{|s|} \sum_{\mathbf{w}_i \in s} \mathbf{w}_i$$

You are given the sentences A, B, and C and the 3-dimensional word vectors $[d_1, d_2, d_3]$ for all occurring words:

- Sentence A: "The **cat** **climbed** the tree."
- Sentence B: "The **feline** **scaled** the tree."
- Sentence C: "The **kitten** **ascended** the tree."

	the	cat	climbed	tree	feline	scaled	kitten	ascended
d_1	0.1	0.4	0.7	1.0	0.35	0.75	0.45	0.77
d_2	0.2	0.5	0.8	1.1	0.55	0.85	0.57	0.87
d_3	0.3	0.6	0.9	1.2	0.65	0.95	0.67	0.97

- (a) Calculate the sentence embeddings $\mathbf{s}_{emb}(A)$, $\mathbf{s}_{emb}(B)$, and $\mathbf{s}_{emb}(C)$ for sentence A, B, and C, respectively, using vector averaging.

Answer

1P

- Sentence A: "The **cat** **climbed** the tree."

$$\begin{aligned}
 \mathbf{s}_{emb}(A) &= \frac{1}{5} (\mathbf{w}_{the} + \mathbf{w}_{cat} + \mathbf{w}_{climbed} + \mathbf{w}_{the} + \mathbf{w}_{tree}) \\
 &= \frac{1}{5} \left(\begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.4 \\ 0.5 \\ 0.6 \end{bmatrix} + \begin{bmatrix} 0.7 \\ 0.8 \\ 0.9 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 1.0 \\ 1.1 \\ 1.2 \end{bmatrix} \right) \\
 &= \frac{1}{5} \begin{bmatrix} 2.3 \\ 2.8 \\ 3.3 \end{bmatrix} = \begin{bmatrix} 0.46 \\ 0.56 \\ 0.66 \end{bmatrix}
 \end{aligned}$$

- Sentence B: "The **feline** **scaled** the tree."

$$\begin{aligned}
 \mathbf{s}_{emb}(B) &= \frac{1}{5} (\mathbf{w}_{the} + \mathbf{w}_{feline} + \mathbf{w}_{scaled} + \mathbf{w}_{the} + \mathbf{w}_{tree}) \\
 &= \frac{1}{5} \left(\begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.35 \\ 0.55 \\ 0.65 \end{bmatrix} + \begin{bmatrix} 0.75 \\ 0.85 \\ 0.95 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 1.0 \\ 1.1 \\ 1.2 \end{bmatrix} \right) \\
 &= \frac{1}{5} \begin{bmatrix} 2.3 \\ 2.9 \\ 3.4 \end{bmatrix} = \begin{bmatrix} 0.46 \\ 0.58 \\ 0.68 \end{bmatrix}
 \end{aligned}$$

- Sentence C: "The **kitten ascended** the tree."

$$\begin{aligned}
 \mathbf{s}_{emb}(C) &= \frac{1}{5} (\mathbf{w}_{the} + \mathbf{w}_{kitten} + \mathbf{w}_{ascended} + \mathbf{w}_{the} + \mathbf{w}_{tree}) \\
 &= \frac{1}{5} \left(\begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.45 \\ 0.57 \\ 0.67 \end{bmatrix} + \begin{bmatrix} 0.77 \\ 0.87 \\ 0.97 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 1.0 \\ 1.1 \\ 1.2 \end{bmatrix} \right) \\
 &= \frac{1}{5} \begin{bmatrix} 2.42 \\ 2.94 \\ 3.44 \end{bmatrix} = \begin{bmatrix} 0.48 \\ 0.59 \\ 0.69 \end{bmatrix}
 \end{aligned}$$

- (b) Which sentence embedding pair is more similar (**A, B**) or (**A, C**)? For your answer, calculate the Manhattan distance D between the embeddings of the sentences in each pair.

The Manhattan distance between two vectors $\mathbf{x}$ and $\mathbf{y}$ is defined as [NLP:III-161]:

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^p |x_i - y_i|,$$

where p is the number of dimensions in the vectors.

Answer

1P

$$\begin{aligned}
 d(\mathbf{s}_{emb}(A), \mathbf{s}_{emb}(B)) &= |0.46 - 0.46| + |0.56 - 0.58| + |0.66 - 0.68| = 0 + 0.02 + 0.02 = 0.04 \\
 d(\mathbf{s}_{emb}(A), \mathbf{s}_{emb}(C)) &= |0.46 - 0.48| + |0.56 - 0.59| + |0.66 - 0.69| = 0.02 + 0.03 + 0.03 = 0.08
 \end{aligned}$$

Answer: The sentence pair (**A, B**) is more similar than (**A, C**) because $D(\mathbf{A}, \mathbf{B}) < D(\mathbf{A}, \mathbf{C})$.

- (c) What are the limitations of the vector averaging approach for generating sentence embeddings? Name at least two.

Answer

0.5+0.5=1P

Any two of:

- It does not capture the word order in the sentence.
- It treats all words in the sentence equally, regardless of their importance or context.
- It does not consider the relationships between words in the sentence.
- It does not account for the semantics of the sentence as a whole.
- It may not perform well on sentences with negation or complex structures.

- (d) Another approach to measuring the similarity between two embedding representations is to calculate the cosine similarity between them [NLP:III-163]:

$$sim_{cosine}(\mathbf{s}_1, \mathbf{s}_2) = \frac{\mathbf{s}_1 \cdot \mathbf{s}_2}{\|\mathbf{s}_1\| \cdot \|\mathbf{s}_2\|} = \frac{\sum_{i=1}^m s_{1,i} \cdot s_{2,i}}{\sqrt{\sum_{i=1}^m s_{1,i}^2} \cdot \sqrt{\sum_{i=1}^m s_{2,i}^2}}$$

(d1) Interpret the following cosine similarity values between two sentence embeddings:

- a) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = -1$
- b) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = 0$
- c) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = 1$

Answer

0P

- a) A cosine similarity of -1 indicates that the two sentence embeddings are in opposite directions in the vector space.
- b) A cosine similarity of 0 indicates that the two sentence embeddings are orthogonal to each other.
- c) A cosine similarity of 1 indicates that the two sentence embeddings are in the same direction in the vector space.

(d2) Calculate the cosine similarity between sentence A and sentence B, and between sentence A and sentence C. Which sentence is more similar to sentence A according to the cosine similarity measure?

Answer

2P (1P for calculations, 1P for answer)

$$\begin{aligned} \text{sim}_{\text{cosine}}(\mathbf{s}_{\text{emb}}(A), \mathbf{s}_{\text{emb}}(B)) &= \frac{\begin{bmatrix} 0.46 \\ 0.56 \\ 0.66 \end{bmatrix} \cdot \begin{bmatrix} 0.46 \\ 0.58 \\ 0.68 \end{bmatrix}}{\left\| \begin{bmatrix} 0.46 \\ 0.56 \\ 0.66 \end{bmatrix} \right\| \cdot \left\| \begin{bmatrix} 0.46 \\ 0.58 \\ 0.68 \end{bmatrix} \right\|}} \\ &= \frac{0.46 \times 0.46 + 0.56 \times 0.58 + 0.66 \times 0.68}{\sqrt{0.46^2 + 0.56^2 + 0.66^2} \times \sqrt{0.46^2 + 0.58^2 + 0.68^2}} = 0.999911 \end{aligned}$$

$$\begin{aligned} \text{sim}_{\text{cosine}}(\mathbf{s}_{\text{emb}}(A), \mathbf{s}_{\text{emb}}(C)) &= \frac{\begin{bmatrix} 0.46 \\ 0.56 \\ 0.66 \end{bmatrix} \cdot \begin{bmatrix} 0.48 \\ 0.59 \\ 0.69 \end{bmatrix}}{\left\| \begin{bmatrix} 0.46 \\ 0.56 \\ 0.66 \end{bmatrix} \right\| \cdot \left\| \begin{bmatrix} 0.48 \\ 0.59 \\ 0.69 \end{bmatrix} \right\|}} \\ &= \frac{0.46 \times 0.48 + 0.56 \times 0.59 + 0.66 \times 0.69}{\sqrt{0.46^2 + 0.56^2 + 0.66^2} \times \sqrt{0.48^2 + 0.59^2 + 0.69^2}} = 0.999992 \end{aligned}$$

Answer: Sentence C is more similar to sentence A according to the cosine similarity measure.
OR They are equally similar (if the students round up the resulting values).